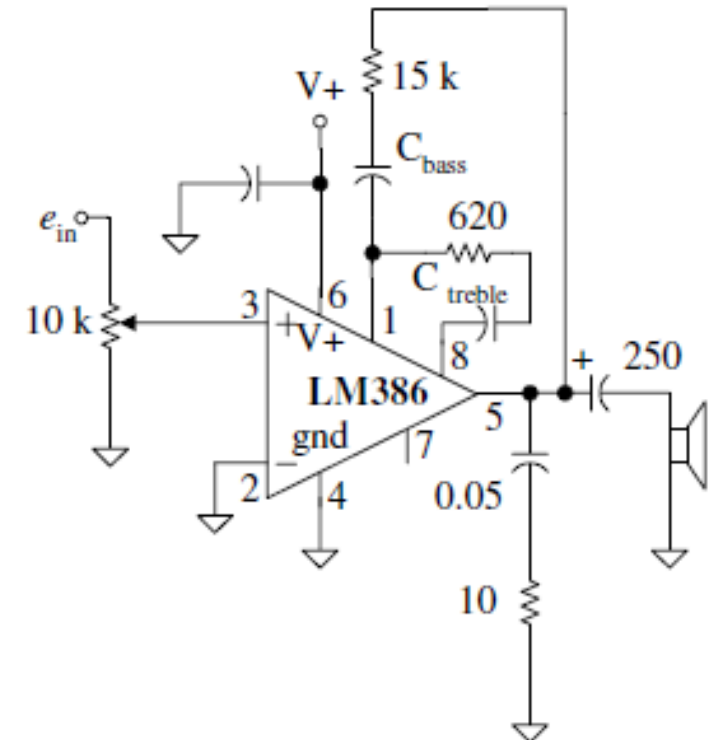


POWER ELECTRONICS

Power Amplifiers

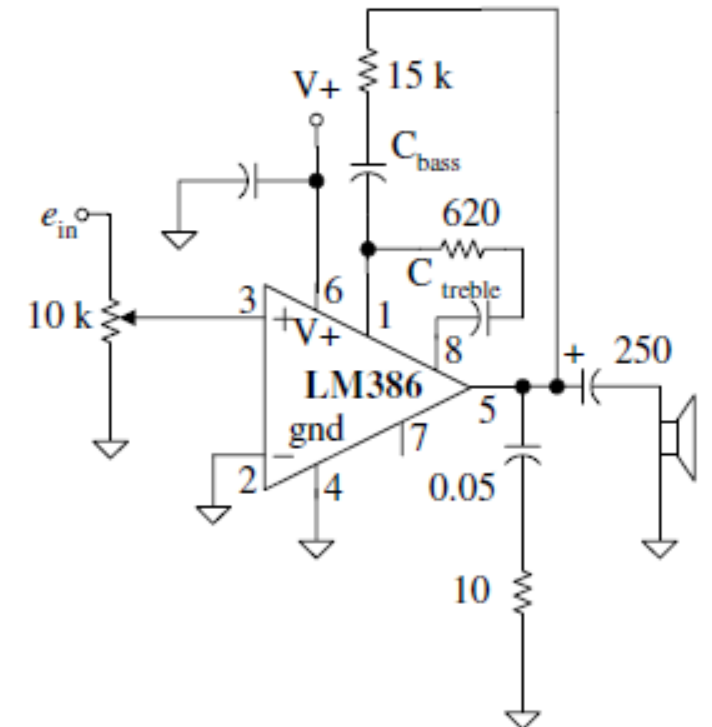
Low Power Audio Amplifier

- LM386 is a low power audio amplifier
- Optimized to require minimum components while delivering as much as 1 W to a loudspeaker.
- Designed to operate from a single power supply
- 4 V to 12 V in most versions
- Biases its output pin to half of whatever voltage is provided as input
- Low frequency cutoff $f_l = \frac{1}{2\pi R_{load} C}$
- Negative feedback is provided internally
- Input impedance is not very large (50 k Ω)
- Negative feedback is internally provided with 15 k between 1 & 5.
- Lower part of feedback voltage divider is 1.35 k between 1 & 8



Low Power Audio Amplifier

- Gain is $A = 2 \frac{15\text{k}\Omega}{R_{pin1-8} + 150\Omega}$
- R_{pin1-8} internally is $1.35\text{ k}\Omega$
- For signal at noninverting pin: gain is 20
- For signal at inverting pin: gain is -20
- To increase the gain, add resistor with dc blocking capacitor between Pin 1 and 8
- If only capacitor is added, gain becomes 200
- Resistor will be parallel to the internal $1.35\text{ k}\Omega$.
- Capacitor's resistance should be small ($\frac{R}{7}$) i.e. $C_{bass} > \frac{1}{2\pi f_1 \frac{R}{7}}$



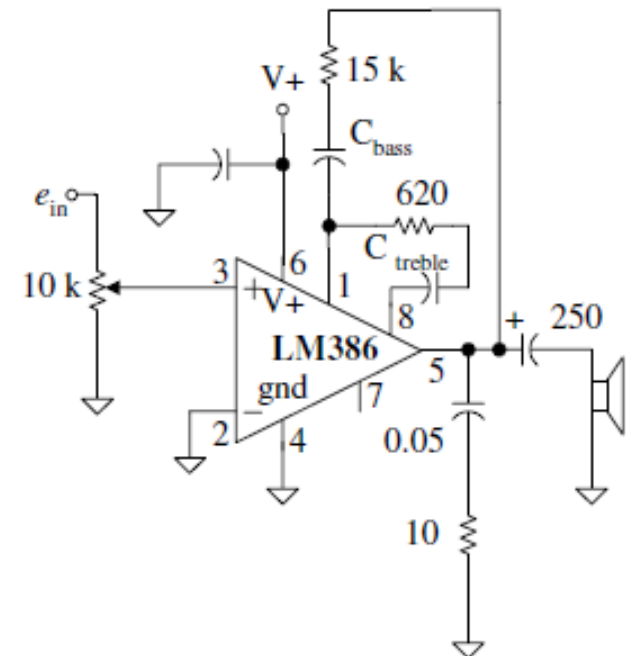
Example

- Determine the resistor and capacitor that must be added between pins 1 and 8 to set the gain at 40, with $f_1 = 80$ Hz

- $620\ \Omega, 27\ \mu\text{F}$

Low Power Audio Amplifier

- Loudness feature: boost the low and high frequency signals
- Bass and treble are the terms used to describe the range of sound in music
- Frequency range in the audio spectrum
- Bass are the low notes (lower end of the human hearing range)
 - Lowest frequency
- Treble are high notes (upper end of the human hearing range)
 - Highest frequency
- Bass and Treble usually get twice gain as that of the middle frequencies



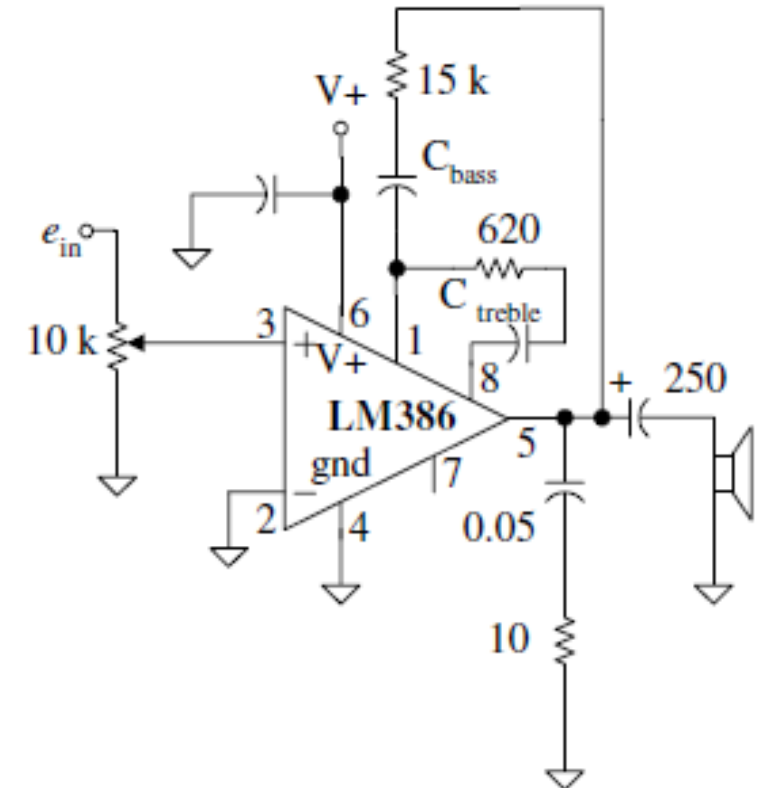
Low Power Audio Amplifier

- Boost the bass by placing series RC between pins 1 and 5
- At low frequencies, capacitor looks like open circuit and Gain is 20
- Internal resistance between 1 and 5 is 15 k
- At high frequencies, capacitor looks like short circuit
 - Placing 15 k resistor between 1 and 5 in parallel to internal resistance
 - Gain becomes

$$A_{mid} = 2 \frac{15k\Omega || 15k\Omega}{1.35k\Omega + 150\Omega} = 10$$

- Lowest stable gain

- Initial value of $C_{bass} \approx \frac{1}{2\pi \times 15k\Omega \times f_{bass\ boost}}$



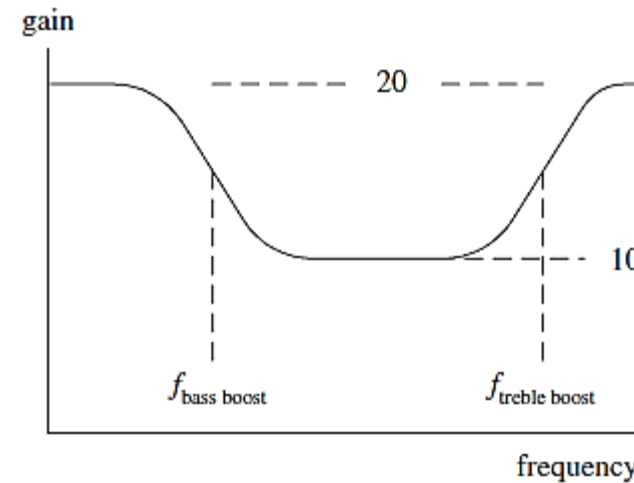
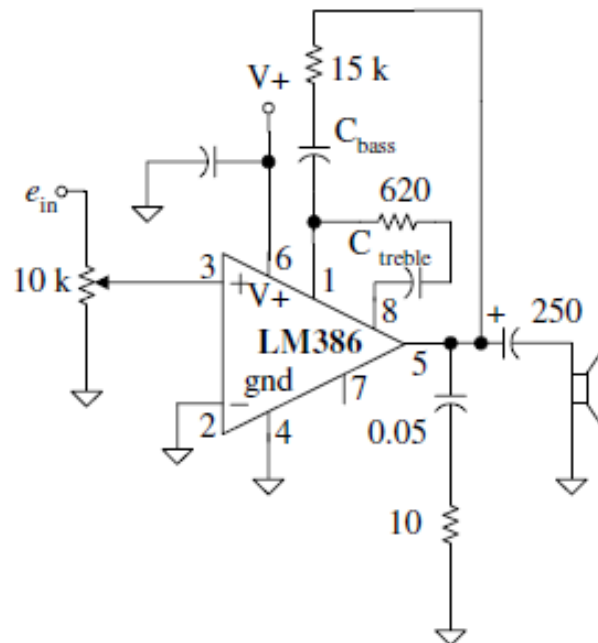
Low Power Audio Amplifier

- For treble boost, select R between pin 1 and 8

$$R_{pin1-8} = 2 \frac{15k\Omega || 15k\Omega}{20} - 150 = 600\Omega$$

- Select the capacitor as:

$$C_{treble} = \frac{1}{2\pi \times R_{treble} \times f_{treble \text{ boost}}}$$



Power Amplifier

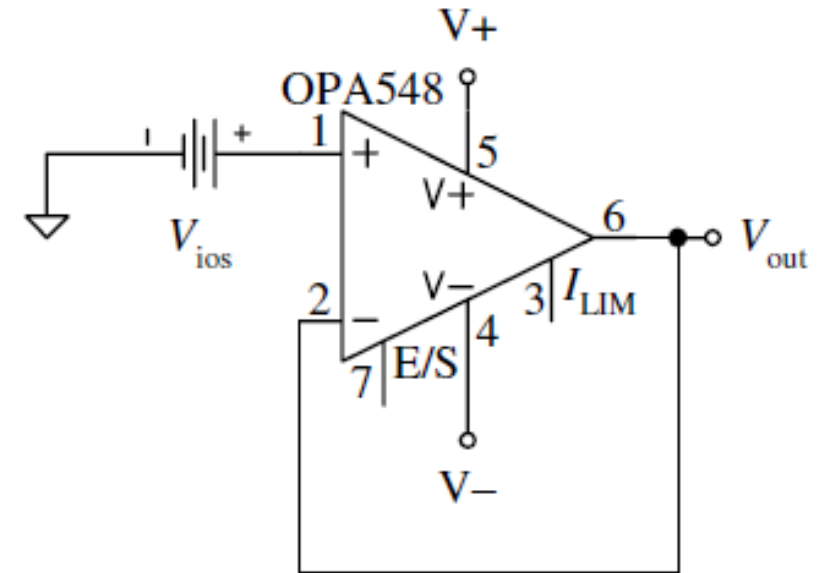
- Convenient way to control power to a load
- Similar to low power ICs but can handle above 50 V at 5 A
- Heat sink
- Thermal protection

OPA548 Operational Amplifier

- Issue 1: In some critical examples this input offset voltage v_{ios} could cause a problem

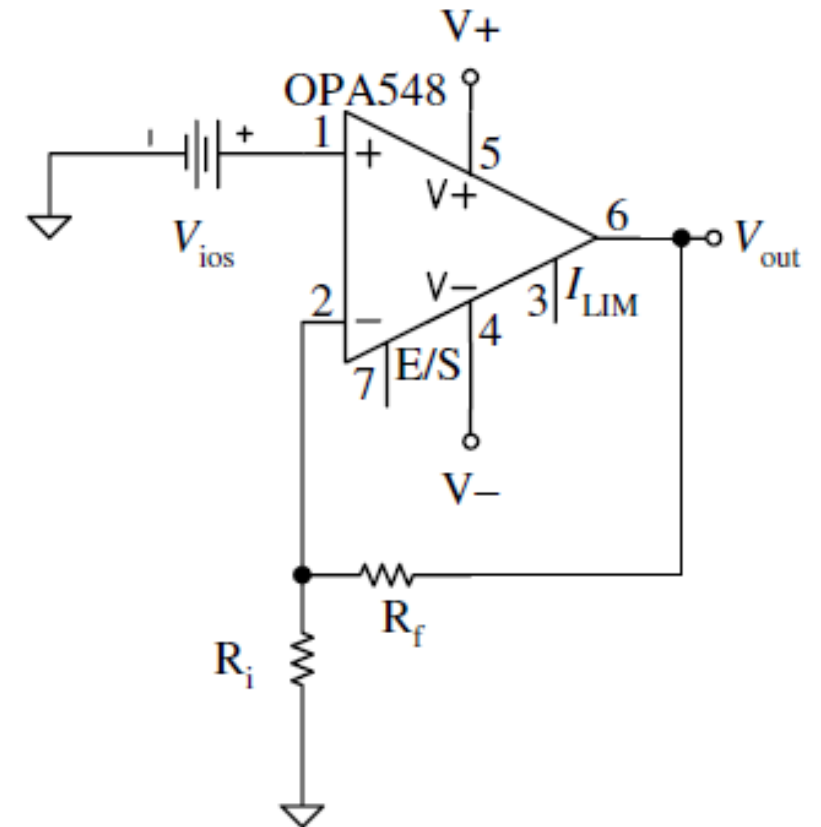
$$V_{out} = \left(1 + \frac{R_f}{R_i}\right) V_{ios}$$

- V_{ios} appears as input voltage that can shift output dc voltage with reasonable gain.
- Coupling capacitor should be connected immediately before the load.
- Issue 2: Input offset current: can be as large as 500 nA



Example 1

- Calculate the output voltage for the circuit in figure shown below with $R_f = 1 \text{ M}\Omega$, $R_i = 10 \text{ k}\Omega$, $V_{ios} = 0 \text{ V}$, $I_{into pin2} = 500 \text{ nA}$
- Repeat the problem with $R_f = 10 \text{ k}\Omega$ and $R_i = 100 \Omega$



Power Calculations with DC Signal

- Simplified circuit to learn concepts

$$P_{load} = V_{load\ dc} \times I_{load\ dc}$$

- Alternate equations for power using Ohm's Law

- Supply power

$$P_{supply} = V_{supply} \times I_{supply}$$

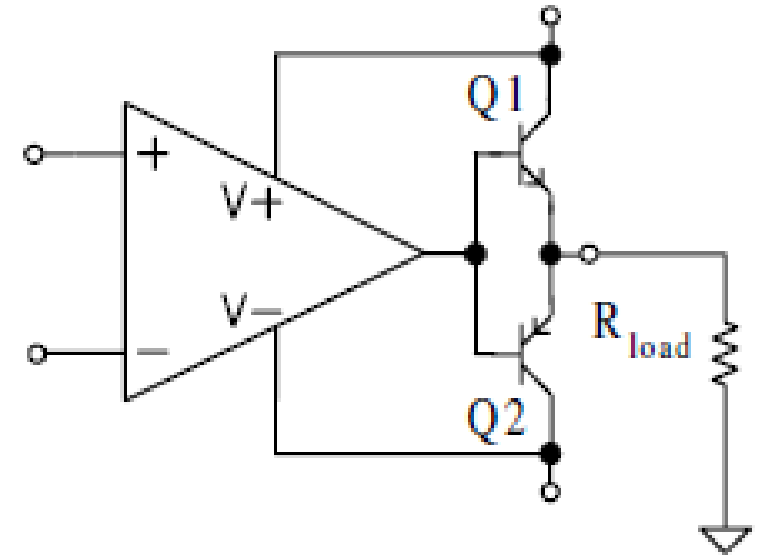
Where $I_{supply} = I_{load\ dc}$

$$P_{supply} = V_{supply} \times I_{load\ dc}$$

- Power dissipation as heat in IC

$$P_{IC} = P_{supply} - P_{load}$$

- Maximum power dissipates in IC when $V_{load} = \frac{1}{2} V_{supply}$



Example 3

- Calculate the power delivered to the load, provided by the supply, and dissipated by the op-amp, for a circuit running from $\pm 12 V_{dc}$, delivering $+5 V_{dc}$ to a 10Ω resistive load.

Power Calculations with Sinusoidal Signal

- Same as done in previous chapter.
- Worst case (maximum dissipation in IC) at $V_{load} = \frac{2}{\pi} V_{supply}$

- Same as done in previous chapter.
- Worst case (maximum dissipation in IC) at $V_{load} = \frac{2}{\pi} V_{supply}$

Example 4

- Calculate the power delivered to the load, provided by the supply, and dissipated by the op-amp for a circuit running from ± 12 V dc, delivering a 5 V peak sine wave to a 10 ohm resistive load.
- What is the effect of increasing the output voltage to 10 V peak?

OPA548 Operational Amplifier

- Gain Bandwidth product is typically 1 MHz (same as 741 amplifier) (for 1 V)

$$GBW = A_o \times f_H$$

Where A_o is the amplifier's closed loop gain, f_H is the high frequency cutoff where the gain has dropped to $0.707A_o$

- For signal above 1 V_P, op-amp's speed is limited by its slew rate (typically 10 V/ μ s) when driving 8 ohm load with a 50 V_{PP} signal.

$$SR = \frac{dv_{out}}{dt}$$

- For triangular wave, slew rate defines the maximum slope of the output, for rectangular output, slew rate sets the smallest rise and fall times. For sinusoidal signal, full power bandwidth is

$$f_{max} = \frac{SR}{2\pi V_P}$$

Example 2

- An audio amplifier has an input of 300 mV_{rms} and is to deliver 35 W to a 8Ω speaker. Can OPA548 be used? Assume the signal is sine wave.

Reference

Power Electronics, Principles and Applications
J. Michael Jacob

Power Electronics
Ned Mohan